PulseIQ Phase 1 Report

Name: Poonam Satish Patole

Track: Data Analysts

Phase: Phase 1

Prepared: 23 October 2026

GitHub repository: PoonamPatole/pulseiq

Git tag: `phase-1`

Task status and evidence

Task	Status	Evidence
1. ETL and data contracts	Complete locally; detailed rules apply to four feeds, and the dictionary supplies type/null checks for all remaining fact columns.	Current ETL results; contract notes
2. Star-schema warehouse	Complete locally; all 12 fact row counts reconcile and 261 relationship exceptions are documented.	Star schema and audit notes; reconciliation results
3. Data-quality gatekeeper	Complete locally; all 14 checks run, each test detects an injected defect, and 11 of 11 planted issue categories were detected.	DQ issue comparison; run `python .\task_3\dq_gatekeeper.py --self-test`
4. Governed metric layer	Complete locally; 28 supplied metrics, M29, catalog, privacy thresholds, and five hand checks are documented.	Hand checks; metric decisions
5. Workforce dashboard	Complete locally; app test passes and screenshots show the KPI cards, M29, trend, department/location charts, and full work-mode chart.	App validation; screenshots in `task_5/evidence/`; CEO questions

Proof it works

1. All 29 CSV files match their `raw` DuckDB row counts.
2. Task 1 rerun verification compared raw, clean, summary, and quarantine contents across two consecutive runs: all 54 materialized tables were stable. The audit run log and DQ result timestamps intentionally change on each run.
3. All 12 warehouse facts reconcile to Task 1 passed counts. Task 1 quarantines 20 bad `fact_task` rows with a written reason; the other fact tables have no quarantined rows.
4. The 14-check gatekeeper detects all 11 planted issue categories. The supplied data currently produces five Critical DQ checks with findings, so the ETL command intentionally exits with code 1 after saving its audit results. This is a data-quality gate failure, not a load crash.
5. The Streamlit app test reports zero exceptions, four KPI cards, four charts, one department selector, and one date-range control. September 2026 headcount is 441 and M29 month-over-month growth is +3.28%. Dashboard screenshots are included in `task_5/evidence/` and in this report.
6. M28 forecast error is `NULL` with `NO_ACTUALS_YET` because actual values for forecast months are not supplied.

What I learned

DuckDB stores a whole analytics database in one local file and can load CSV data directly.

A star schema keeps measurable events in fact tables and descriptive context in dimensions.

An employee surrogate key preserves which department was valid when a historical event happened.

Data contracts and quality checks make failures visible instead of silently hiding them.

Metric views give dashboards one reusable definition for each business number.

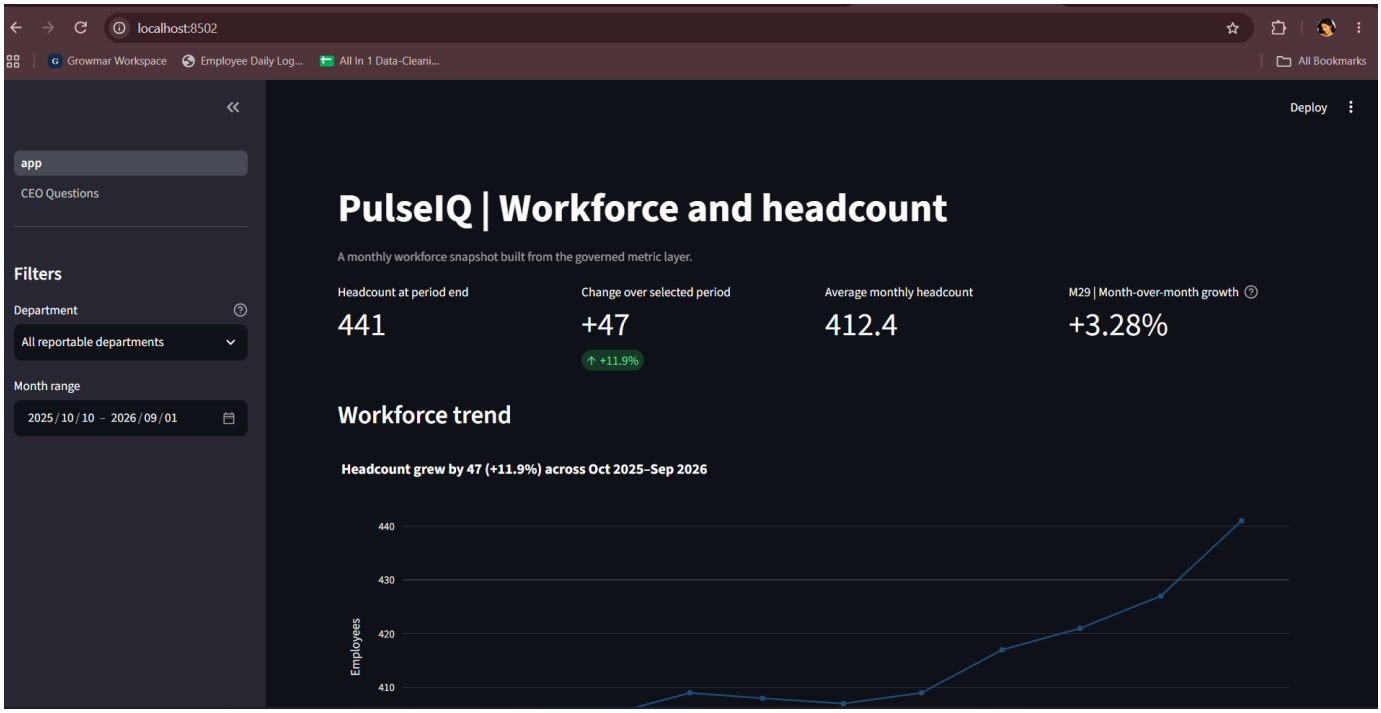
Problems and solutions

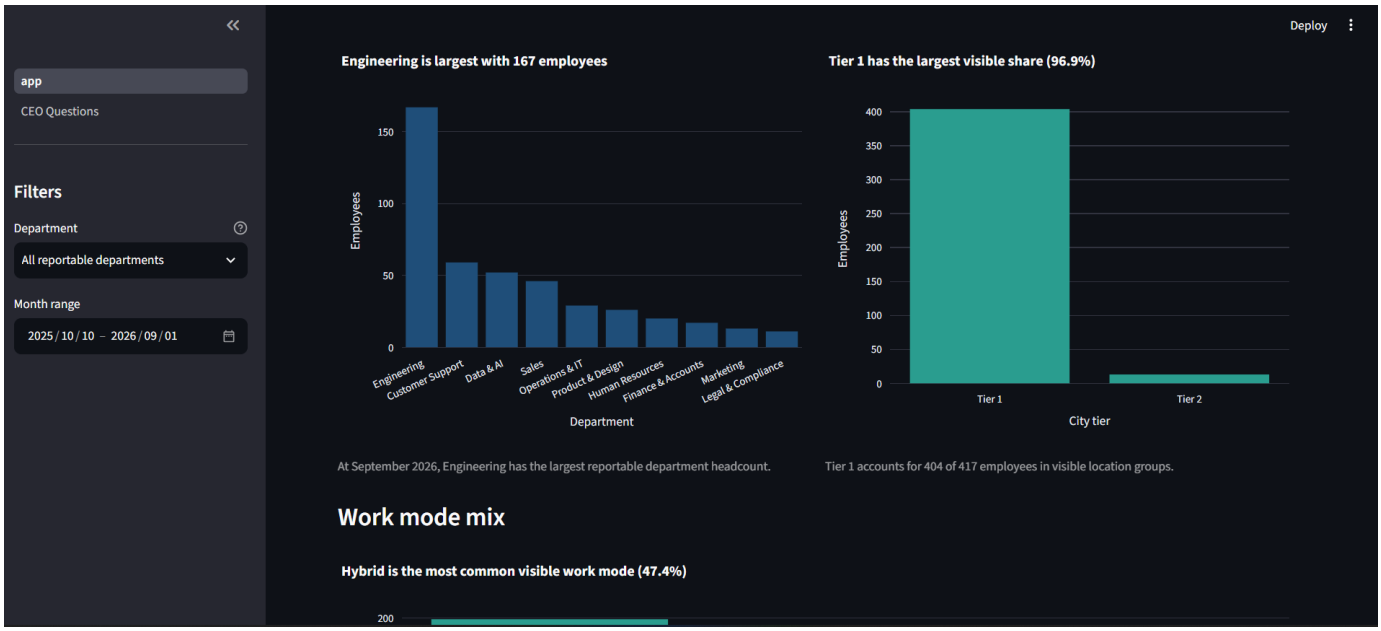
1. The starter contract sheet defines detailed rules for only four feeds. I applied the provided data dictionary's type and nullability to every fact column, and did not invent allowed-value rules for the remaining feeds.
2. CSV type inference initially treated numeric-looking text identifiers as numbers. The pipeline converts those values only in `clean`; the raw source remains unchanged.
3. The quality gate finds existing payroll, key, history, and minimum-group-size issues. They are recorded in `audit.dq_results` and quarantines rather than removed silently.
4. The dataset has 261 explainable relationship exceptions: employee history date gaps, forecast dates beyond the supplied calendar, and two payroll rows with employee key `-1`.
5. The saved dashboard screenshots show the KPI row, trend, filters, and workforce breakdown charts.

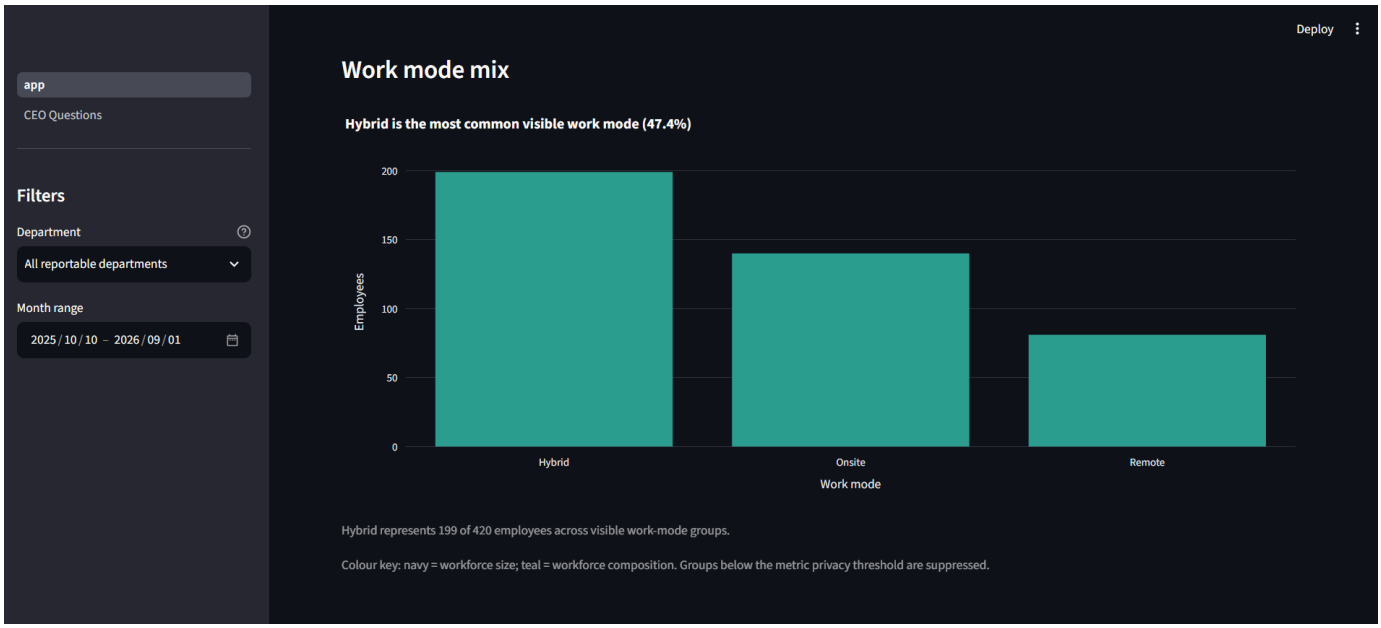
Plan for Phase 2

Resolve or formally accept the Critical data-quality findings with documented owners. Then extend the dashboard to attendance, attrition, teams, and payroll cost, using the existing governed metric views and preserving small-group privacy thresholds.

Dashboard screenshots






Deploy
app
CEO Questions
Filters
Department
All reportable departments
Month range
2025 / 10 / 10 – 2026 / 09 / 01
Work mode mix
Hybrid is the most common visible work mode (47.4%)
200
150
100
50
0
Employees
Hybrid
Onsite
Remote
Work mode
Hybrid represents 199 of 420 employees across visible work-mode groups.
Colour key: navy = workforce size; teal = workforce composition. Groups below the metric privacy threshold are suppressed.